4 **(a)** With reference to molecular kinetic and potential energies, explain why the internal energy of the system increases when a wire is stretched within its elastic limit at constant temperature.

.....

.....

.....

..... [2]

(b) A fixed mass of an ideal gas is initially at a temperature of 17 °C.

The gas has a volume of 0.24 m^3 and a pressure of 1.2×10^5 Pa.

(i) Calculate the number of moles n of the gas.

n = ... mol [1]

(ii) The gas undergoes three successive changes, as shown in Figure 4.1.

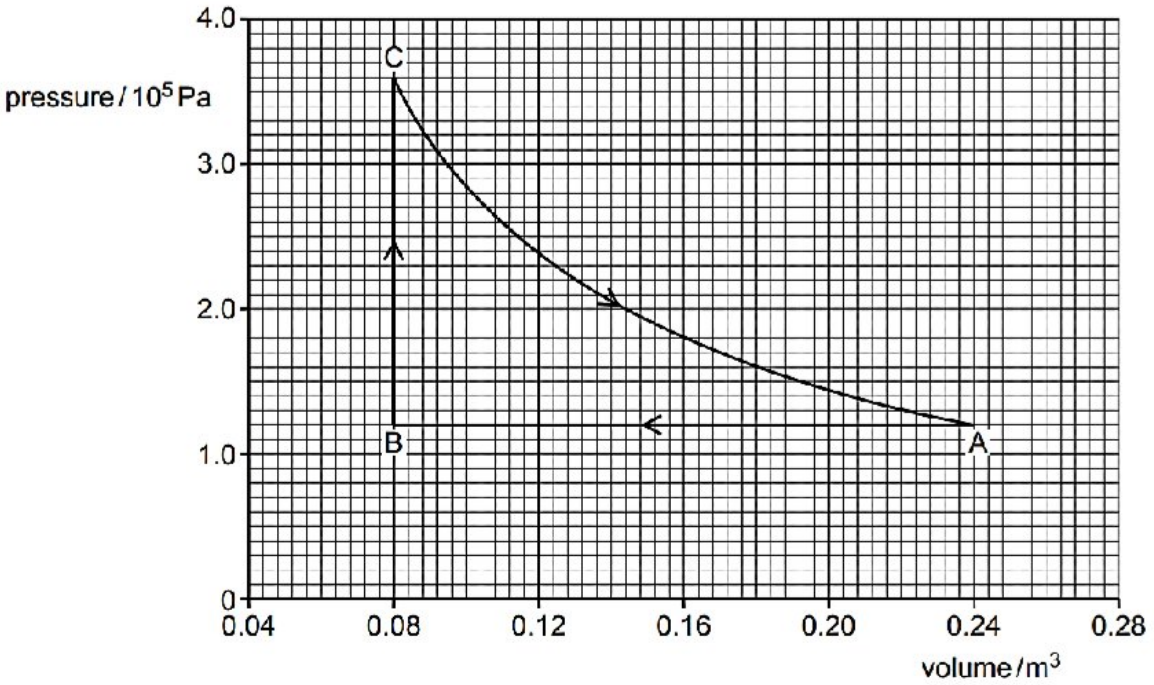


Figure 4.1

The initial state is represented by point A. The gas is cooled at constant pressure to point B by the removal of 48.0 kJ of thermal energy.

The gas is then heated at constant volume to point C.

Finally, the gas expands at constant temperature back to its original pressure and volume at point A. During this expansion, 31.6 kJ of work is done.

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Complete Table 4.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes AB, BC and CA.

change	work done on gas / kJ	thermal energy supplied to gas / kJ	increase in internal energy of gas / kJ
AB		–48.0	
BC			
CA			

Table 4.1

[3]

[Total: 6]